Database Forensics

How Data Solves Crimes in Large Organisations

A 7-slide guide to investigating data at scale with SQL and audit logs

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DATABASE FORENSICS = DIGITAL EVIDENCE

APPLICATION EXERCISES

QUESTIONS:

1. When an HR or DBA uses their access to DROP tables, OVERWRITE codes and ALTER schemas to hide fraud. What footprints are physically impossible to erase?

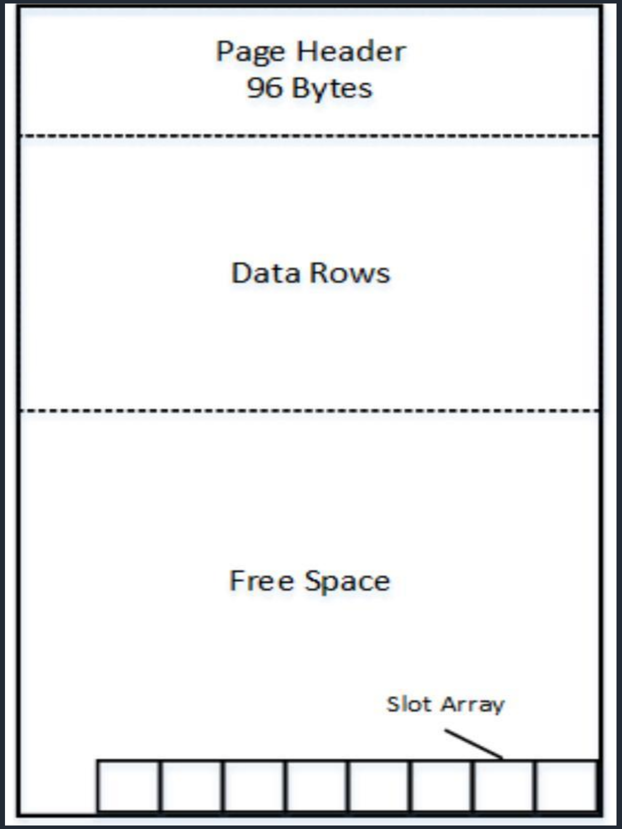
2. If a ghost worker looks 100% clean inside payrolls, how do you prove that they don’t exist in the real world using multi-layered data analytics?

3. Which method is most effective MANUAL or DATABASE FORENSICS?

SQL + Audit Logs + Timestamps = The Truth

THE ANATOMY OF DATABASE PAGE (WHY DATA LINGERS)

Database doesn’t write data randomly; they store it in structured blocks called pages.



PAGE HEADER

(System notes: Page ID, space left, etc.)

DATA ROWS

[Row 1: Admin logged In]

[Row 2: Fraudulent paycheque sent to account x] ⟵ What the suspect wants gone

[Row 3: System Normal]

FREE SPACE

(Empty room waiting for new data)

SLOT ARRAY

[Pointer 1: -> Row 1]

[Pointer 2: -> Row 2] X (change to “empty space”) ⟵ The shortcut

[Pointer 3: -> Row 3]

HERE,

WAL(Write-Ahead Logging)

TRANSACTION LOG
Timestamp: 10:14:02 AM
User Account: HR_Officer_01
Action: Executed DELETE command on Table ‘Payroll’ for Row ID 942

Active Directory (AD) Check

The physical Access test(badge swipes or biometric logs)

SQL Banking Interrogation (The Cash Routing Trick)

FORENSIC TAKEWAY: Unless a drive is physically shredded or magnetically wiped, remnant survive. Between raw data carving from unallocated space and reconstruction history via Write-Ahead Logs(WAL) , an investigator can almost always rebuild the timeline and find the missing file.

- True verification happens at the intersection of disconnected silos.
- Structural destruction does not equal data destruction; even sophisticated data manipulation cannot erase a digital footprint.
- While the structural pointers are gone, the raw data pages often remain intact in the database files until maintenance routines or page allocations reuse that space.

SOLUTIONS

QUESTION ONE: THE ROGUE ADMINISTRATOR

- CHECK what is still in the DB
- Soft delete (delete_at = now() the row is still there.
- Hidden in history table: look for audit/ history tables.(employee_history, audit_log, pg_log, sys.audit.)
- Indexes: sometimes the index entry stays longer than the row.

- Use transaction logs,
- Postgres: WAL files in pg_wal/. Tool like pg_waldump can read them.
- MYSQL/MariaDB: Binlogs and redo logs. mysqlbinlog lets you see INSERT/DELETE statements.

- SQL Server: Transaction log LDF files. Tools like fn_dblog can read them.
- Oracle: Redo logs and archive logs.

- Check backups and snapshots(most companies have daily backups, snapshots, or point - in - time recovery.
- File System level recovery(if the actual database file was deleted from disk :

- DON'T write on the disk. Stop the DB and unmount the volume.
- Use file carving tools: scalpel, foremost, autopsy to recover deleted.dbf, .mdf, .ibd files.

- Look at file system metadata: MFT in NTFS, inode in ext4. it tell you when the file was deleted.
- Memory and page files(running database keep data in RAM and buffer pools.

FORENSIC RECOVERY TECHNIQUES

- Log Analysis
- File Carving
- Page Scraping

- Git History Carving
- (memory caches, temp directories, or unallocated space on the web/application server

- The Timestamped Discrepancy

- Hex Editor and Disk Imaging Tools
- Encase/FTK Imager
- 010 Editor/ WinHex

- ApexSQL Recover/ Quest Software

- SysTools SQL Recovery

- FQLite/ Undark

SQL Banking Interrogation

QUESTION 2: THE GHOST WORKER DETECTION

- ❑ Check login and activity logs(IDs, IP addresses, timestamps(payroll records but 0 activity
- ❑ Check Audit Trails(created,modified,odd hours)
- ❑ Check payroll Payouts(match employee ID to bank account
- ❑ RED FLAG = multiple employee Same account, or account doen’t match HR data

- ❑ Payroll DB vs HR DB vs LDAP vs Email server vs Banking payout file.
- ❑ “Impossible “ state: 2 people with same SSN, 1 bank account for 1 employees, salary paid but no tax deduction.
- ❑ Look at backups and snapshots. attackers rarely delete all of them.

Can we prove every paid employee exists in HR and IT systems?

Who has DBA access, and are their actions logged outside the DB?

Do we test our backups by restorng them?

Manual checks catch obvious fraud. Cross-layer forensics catches hidden fraud.

Database Forensics is how an organization protects its entire structural intergrity.

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